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BSH GUIDELINE

Diagnosis and monitoring of thrombotic thrombocytopenic purpura: Guidance for laboratories from the British Society for Haematology

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Summary
This guideline complements previous guidelines, focussing on the laboratory methods associated with testing for deficiency of the metalloproteinase ADAMTS13.

KEYWORDS
ADAMTS13, laboratory guidance, thrombocytopenia with haemolysis, thrombotic thrombocytopenic purpura

METHODOLOGY

This guideline was compiled according to the British Society for Haematology (BSH) process at [https://b-s-h.org.uk/media/or2db2zv/2024-08-20-bsh-guidelines-development-process.pdf]. The Grading of Recommendations Assessment, Development and Evaluation (GRADE) nomenclature was used to evaluate levels of evidence and to assess the strength of recommendations. The GRADE criteria can be found at https://www.gradeworkinggroup.org/ and is summarised in appendix 3 of the guidance document linked above.

Review of the manuscript

Review of the manuscript was performed by the BSH Haemostasis and Thrombosis Task Force and the BSH Guidelines Committee. It was also on the members section of the BSH website for comment.

INTRODUCTION

This document is intended as a guide for laboratories in the United Kingdom that may receive requests for assays for the metalloproteinase ADAMTS13 and covers aspects of testing that should be considered by all haematology laboratories. In the United Kingdom, assays for ADAMTS13 activity, anti-ADAMTS13 antibodies and ADAMTS13 antigen for the diagnosis or exclusion of thrombotic thrombocytopenic purpura (TTP) are usually performed in regional TTP centres. However, it is recognised that most haematology laboratories receive occasional requests for assays for ADAMTS13, even if they are not regional TTP centres.

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LABORATORY INVESTIGATION FOR TTP

Suspected TTP is a medical emergency where availability of an ADAMTS13 activity is not a prerequisite for immediate management or treatment.[1] It has a prevalence of six to ten cases/million,[2] and an annual incidence of approximately one new case/million,[3] similar to the incidence of acquired haemophilia A. Most cases are immune-mediated TTP (iTTP), with <5% of presenting cases being due to congenital TTP (cTTP).[4]

Laboratory suspicion of TTP should be aroused when there are features of microangiopathic haemolytic anaemia and thrombocytopenia, especially if the platelet count is $<30\times10^9$/L. If a full blood count identifies a platelet count $<30\times10^9$/L, a blood film should be examined as soon as possible, ideally within 60 min, and if red cell morphology is suggestive of haemolysis, this should be referred for immediate review by a clinical haematologist. Appropriate additional investigations for confirmation of in vivo haemolysis should be considered. A high reticulocyte count ($>120\times10^9$/L), undetectable serum haptoglobin and an elevated lactate dehydrogenase level may also be seen.[5] The presence of schistocytes on a blood film is a hallmark of the disease, but these are not always detected.[6] Routine coagulation parameters are usually normal.

As laboratory findings of thrombocytopenia and haemolysis are not specific for TTP and may be present in disseminated intravascular coagulation (DIC) and in other thrombotic microangiopathies (TMA), including haemolytic uraemic syndrome, investigations should include urgent analysis of ADAMTS13 activity when there is clinical suspicion of TTP.

A suggested workflow is shown in Figure 1.

Recommendations

- Haematology laboratories should have documented processes in place to ensure that new patients with full blood count samples with thrombocytopenia, especially those with platelet count $<30\times10^9$/L, are examined microscopically for evidence of haemolysis. (Grade 1C)
- Haematology laboratories should have a documented process in place to ensure that new patients with blood films with thrombocytopenia and evidence of haemolysis are escalated rapidly for clinical review. (1C)

TURNAROUND TIMES

Failure to promptly diagnose TTP can cause delay in treatment and has significant clinical consequences. A UK-wide multicentre audit[7] highlighted that diagnostic uncertainty was the most common reason for delay, defined as more than 8 h from the first full blood count to issue of plasma for plasma exchange (PEX) for treatment-initiation in acute TTP. This delay was more frequent in patients transferred to tertiary centres. To address this, NHS England commissioned nine regional TTP centres (11 laboratories); specialist testing is performed in regional TTP centres in Scotland, Wales and Northern Ireland. These regional centres follow a clear referral pathway to ensure timely treatment and access to ADAMTS13 assays.

Prompt assaying of ADAMTS13 activity at diagnosis not only identifies patients with TTP who may benefit from ongoing treatment, but also significantly reduces the incidence of unnecessary treatment and exposure to blood products during PEX for patients without TTP,[8] with reduced expenditure on plasma and associated resources.

Rapid automated assays deliver quantitative ADAMTS13 activities in 30–40 min, with good performance in verification studies,[9] enabling prompt diagnosis and reducing reliance on empirical treatment and improving resource allocation.

Health-economic studies consistently support rapid testing, as early differential diagnosis reduces the demand for PEX and caplacizumab.[10–14]

Recommendations

- Haematology laboratories should have access to urgent ADAMTS13 activity testing either via sample transfer to the laboratory of a regional TTP centre, or on-site. (1B)
- Local protocols should be established for urgent transfer of pre-treatment samples for ADAMTS13 activity assays in patients with suspected TTP. (1C)
- In cases of suspected TTP where an urgent ADAMTS13 activity assay is required, preliminary results should be available as soon as possible, but ideally within one hour of sample receipt by the testing laboratory for laboratories with rapid tests available, or six hours for laboratories using enzyme-linked immunosorbent assay (ELISA). (1C)

PRE-ANALYTICAL CONSIDERATIONS FOR ADAMTS13 TESTING

Clinical considerations

When TTP is suspected, samples should be collected prior to treatment.[1,15] Following a clinical discussion between haematologists at referring sites and TTP centres, the laboratory of the regional TTP centre must be contacted to confirm imminent dispatch of the sample and be notified of the urgency of the request. Contact details for clinicians and laboratories at regional TTP centre laboratories in England can be found at https://www.nationalttpgroup.org/specialist-centres/.

Requests for monitoring of known TTP patients should be assessed by the requesting clinician to determine if urgent processing is required.

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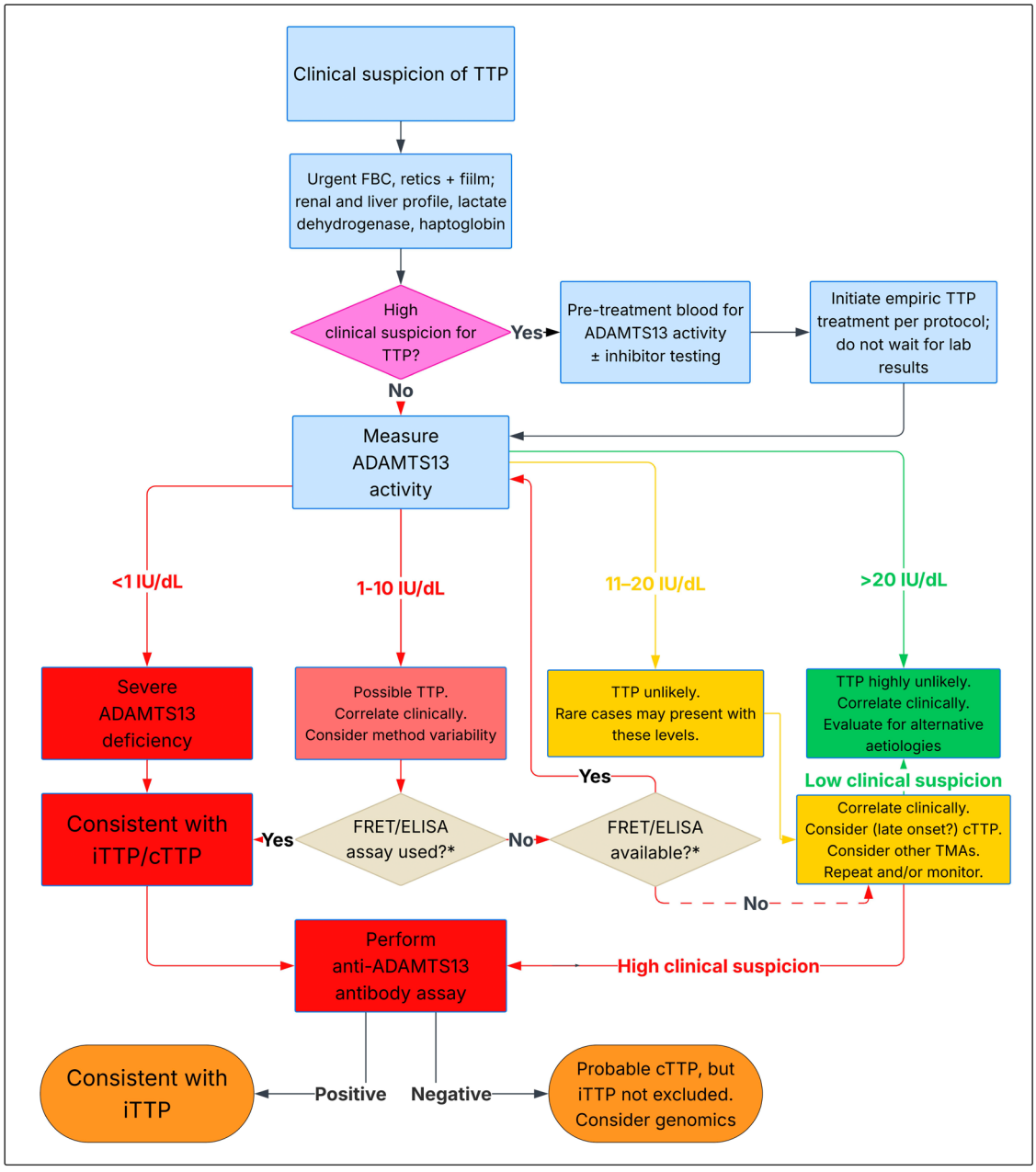


FIGURE 1 Suggested pathway for laboratory testing in patients with suspected TTP. *Some automated ADAMTS13 activity assays may be found to be comparable to FRET/ELISA assays following further clinical evaluation. ELISA, enzyme-linked immunosorbent assay; FRET, fluorescence resonance energy transfer.

Sample collection, preparation and transport

Samples should be collected into sodium citrate 3.2% (0.109 M), ensuring a fill of at least 90%, and mixed by gentle inversion. Underfilled or overfilled samples are not acceptable for use.[16] Citrated whole blood must be transported to the laboratory at ambient temperature 18–25°C[17] with minimal delay. Whole blood samples must be transported to a laboratory at a regional TTP centre at ambient temperature immediately.

Plasma samples should be prepared by centrifugation at 18–25°C at 2000 *g* for 15 min to ensure platelet depletion.[16] If not for immediate testing, samples should be stored frozen in a screw-capped polypropylene tube with an 'O'-ring below −24°C (ideally below −70°C), as previously described[16]; frozen plasma samples being transported to a laboratory at a regional TTP centre must be transported on dry ice.[16] In rare cases where treatment has started in a patient and a pre-treatment sample is available at the referring site, it can be used to establish pre-treatment ADAMTS13 activity as long plasma is frozen within 48 h of sample collection. Samples that thaw in transit should be rejected for analysis; for this reason, it is recommended that

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additional frozen plasma aliquots are stored below −70°C by the requesting laboratory. Additional aliquots may also be required for ADAMTS13 antibody assays if ADAMTS13 activity is <10 IU/dL.

Serum samples may be used in circumstances where a citrate sample is not available (as advised by a regional TTP centre), but samples collected into ethylenediaminetetraacetic acid (EDTA) cannot be used as EDTA results in falsely low ADAMTS13 activity.[15]

Sample processing

Frozen samples should be thawed in a water bath at 37°C for 5 min (or until fully thawed)[16,17] and tested as soon as possible. Thawing for longer periods is not recommended as this can lead to reduced ADAMTS13 activity.[18]

Standardised information required by laboratories at regional TTP centres

All regional TTP centres have a request form available for ADAMTS13 assays. If such a form cannot be found in time to be sent with a sample, the following information must be provided with every request:

- Patient's details (name, DOB, medical record number and/or NHS number);
- Date and time of sample collection;
- Clinical details;
- Contact details for consultant haematologist at the referring site;
- Return address and telephone number for return of results;
- 'Has treatment and/or patient transfer been considered?' (if known); and
- Statement of urgency.

Failure to provide these details may lead to delays in processing of samples or reporting of results, especially if the sample arrives at the reference laboratory without a prior telephone call.

Laboratory and clinical contact details for samples being sent to regional TTP centre laboratories in England can be found at https://www.nationalttpgroup.org/specialist-centres/.

Recommendations

- Samples should be collected into 0.109 M sodium citrate. (1B)
- For patients with a strong clinical suspicion of TTP, samples must be sent to a regional TTP centre immediately if the patient has not already been transferred. (1B)

TESTING FOR ADAMTS13 ACTIVITY

ADAMTS13 activity assays measure the ability of patient plasma to degrade/cleave an external source of von Willebrand factor (VWF). Refer to Table 1a.

Quantitative ADAMTS13 activity assays

First-generation assays for ADAMTS13 activity use full-length unmodified VWF, with a wide variety of methods used to measure VWF multimeric size at the end of testing. Extended pre-test treatment of VWF is required to make it accessible for cleavage, resulting in multi-day turnaround times for these assays. Such assays are now rarely used in clinical settings, and some may be affected by caplacizumab.[26]

Second-generation activity assays use truncated VWF fragments that can be more readily cleaved by ADAMTS13. These VWF fragments contain a section of VWF A2 domain, including the ADAMTS13 VWF-cleavage site at Tyr1605-Met1606, and key sites of VWF and ADAMTS13 interaction. The most commonly used VWF fragment is VWF73, a 73 amino acid sequence of the A2 domain (Asp1596-Arg1668).[27]

Manual second-generation assays are commonly used (refer to Table 1a); they have a turnaround time less than 5 h, but their use is typically limited to specialist sites. The VWF73-fluorescence resonance energy transfer (VWF73-FRET) method measures cleavage of the fragment directly through the separation of a fluorophore and a fluorophore-quencher attached either side of the cleavage site, with separation resulting in a fluorescent signal. The Technozyme ADAMTS13 activity two-step ELISA (Technoclone GmBH, Vienna, Austria) is widely used and measures cleavage of VWF indirectly using a monoclonal antibody (N10), which recognises and binds to a novel epitope revealed when VWF73 is cleaved by ADAMTS13.[19] Conjugate and substrate are then used to detect bound monoclonal antibody (N10), with a chromogenic signal being proportional to ADAMTS13 activity.

Two automated ADAMTS13 activity assays are currently commercially available in the United Kingdom; other assays have shown good results in small studies but are not yet available[25,28] (refer to Table 1b). The Technofluor ADAMTS13 activity assay (Technoclone GmbH, Vienna, Austria) is performed on the Technoclone Ceveron s100[24] and is a direct (FRET) assay using the same VWF73-FRET technique used in the manual FRET method. The HemosIL ADAMTS13 activity assay (Werfen/Instrumental Laboratory, Bedford MA, USA) on the Werfen Acustar/Bio-Flash platform is an indirect chemiluminescence enzyme immunoassay (CLEIA), using the same VWF73 molecule and N10 antibody as the Technozym ELISA.[22,23] Direct (FRET) assays measure VWF73-cleavage within the reaction solution, while current indirect assays (ELISA and HemosIL CLEIA) measure VWF73 attached via a matrix, with the reaction solution washed away.

While there are similarities in assay principle and materials among second-generation VWF73 assays, there

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TABLE 1 Summary of known testing platforms used in clinical laboratories for ADAMTS13 activity and antigen assays.

(a) Manual ADAMTS13 activity assays		
Method	**Detection and utility**	**VWF source**
TECHNOZYM ADAMTS13 Activity[19]	Quantitative ELISA	VWF73-GST
IMMUCOR ATS-13 Activity	Quantitative ELISA	VWF73-FRET
ACTIFLUOR ADAMTS13 Activity	Quantitative ELISA	VWF86-Alexa FRET
Laboratory-developed Tests	Quantitative ELISA	VWF73-FRET
TECHNOSCREEN ADAMTS13[20,21]	Semi-quantitative flow through assay	VWF73-GST

(b) Automated ADAMTS13 activity assays			
Method	**Detection and utility**	**VWF source**	**Analyser platform**
HemosIL AcuStar ADAMTS13 Activity[22,23]	Quantitative CLEIA	VWF73-GST	Werfen AcuStar/Bio-Flash
TECHNOFLUOR ADAMTS13 Activity[24]	Quantitative CLEIA	VWF73-FRET	Technoclone Ceveron s100
Sysmex ADAMTS13 Activity[25]	Quantitative CLEIA	Biotinylated-VWF73	Sysmex CN-3500/6500

(c) ADAMTS13 antigen assays	
Method	**Detection**
TECHNOZYM ADAMTS13 Antigen	Quantitative ELISA
TECHNOZYM ADAMTS13 Activity/Antigen	Quantitative fluorogenic ELISA
IMUBIND ADAMTS13	Quantitative ELISA
Laboratory-developed Tests	Quantitative ELISA

Note: Data from the ECAT Foundation (unpublished data, personal communication).

Abbreviations: CLEIA, chemiluminescent enzyme immunoassay; ELISA, enzyme-linked immunosorbent assay; FRET, fluorescence resonance energy transfer; GST, glutathione S-transferase.

is variability in the constituents of buffers and diluents, dilution factors, incubation periods, calibration materials and whether the VWF73 is measured in the reaction solution or matrix, all of which may impact direct method comparability.

All current versions of major international guidelines state that ADAMTS13 activity <10 IU/dL confirms the diagnosis of TTP, while International Society on Thrombosis and Haemostasis (ISTH) guidelines also include an equivocal range of 10–20 IU/dL where results are not considered negative but other diagnoses should be considered.[1,15,29–30] Sensitivity/specificity of 97%/100%[1] and 94%/99%[29] using ADAMTS13 activity <10 IU/dL for confirmation of TTP were established from studies using the manual VWF73-FRET assay.[1,29,31] The cut-off value includes a range that captures cases of cTTP.[32] Differences between direct and indirect assays may influence variability in sensitivity and specificity across the clinically relevant range.[33–35]

Unpublished data from the UK TTP Registry found that 93% of patients with iTTP in the last 5 years presented with no detectable ADAMTS13 activity levels (<1 IU/dL). However, some patients with iTTP present with ADAMTS13 activity levels between 1 and 10 IU/dL.[36] Patients with cTTP show greater variability in activity levels at presentation; ADAMTS13 activity may be 5–12 IU/dL[37] in patients with the p.(Arg1060Trp) *ADAMTS13* variant, the most common variant[38] seen in the United Kingdom. All assays described above demonstrate high sensitivity and specificity for the diagnosis of TTP at very low activity levels, but there are grey areas, where cases of iTTP are less likely and rare cases of non-TTP in patients with low ADAMTS13 activity occur. For the manual FRET and ELISA, there is a grey area for ADAMTS13 activity between 10 and 20 IU/dL where result interpretation requires clinical expertise and the context of the patient presentation. A comparable grey area in the current automated platforms has not been established. Particular attention must be paid to the current HemosIL CLEIA assay, which has been reported to produce erroneously low ADAMTS13 activity results (<10 IU/dL) in diagnostic samples from patients that have given results above 10 IU/dL by other assays.[33,34] It remains to be seen whether future automated assays[25,28] address erroneous results, but a grey area between different assays is likely to remain. Pre-test screening and exclusion of inappropriate requests[39,40] may reduce this occurrence, but care needs to be taken that testing is not delayed in low probability requests from young patients who may have cTTP,[38] nor from elderly patients with iTTP in whom clinical scoring is less effective.[41]

Despite these variations, clinical guidelines maintain that a cut-off of 10 IU/dL is used.[1,15,29–30] Early confirmation with an alternative assay should be obtained in all instances when ADAMTS13 activity does not give results consistent with pre-test probability and clinical assessment; a repeat sample may not aid diagnosis if the same assay is used. Demonstration of an anti-ADAMTS13 antibody can confirm a diagnosis of iTTP.

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TABLE 2 Summary of known testing platforms used in clinical laboratories for anti-ADAMTS13 inhibitor assays.

Method	Detection and utility	Key method details	Interpretation of results
TECHNOZYM ADAMTS13-INH	Quantitative ELISA for neutralising and non-neutralising IgG-antibodies	Full-length recombinant ADAMTS13 for antibody capture Calibrated using lyophilised plasmas with anti-ADAMTS13 Ab from 0 to ~100 arbitrary units (U/mL) traceable to an in-house standard	<12.0 U/mL: Negative 12–15 U/mL: Borderline >15 U/mL: Positive
Laboratory-developed test	Quantitative ELISA for neutralising and non-neutralising IgG-antibodies	Full-length recombinant ADAMTS13 for antibody capture Calibrated using a high titre anti-ADAMTS13 Ab patient plasma (defined as 100%)	<6.1%: Negative ≥6.1%: Positive
Bethesda-style assay	Quantitative assay for neutralising antibodies	Incubate heat-treated patient plasma with normal pooled plasma. Normal pooled plasma added to doubling-dilutions of heat-treated patient plasma diluted in buffer for 30 min to 2 h Residual ADAMTS13 activity measured by in-house ADAMTS13 activity assay	<0.5 BU/mL: Negative ≥0.5 BU/mL: Positive

Semi-quantitative screening assay

The semi-quantitative Technoscreen assay (Technoclone GmbH, Vienna, Austria) has a high sensitivity and negative predictive value that may make it effective for ruling out iTTP,[20,21] but reports of false negatives[42] mandate confirmation with a quantitative ADAMTS13 activity assay for all results.

Considerations when introducing an assay

New platforms for ADAMTS13 activity assays are becoming increasingly available, and testing has become possible outside regional TTP centres. However, when considering introduction of a new ADAMTS13 activity assay in any centre, laboratories and clinicians must have a clear understanding of the expected ranges in iTTP and cTTP at presentation, and in iTTP relapse, and should carefully consider whether testing locally may contribute to unacceptable delays in appropriate treatment for TTP. There must be clear clinical pathways for dealing with both normal and abnormal results.

While there are no specific requirements for validating or verifying ADAMTS13 assays in addition to published guidance,[15,43–44] samples selected must reflect local requesting patterns so that patients with other suspected TMAs, including those with low clinical probability of TTP, are represented.

Recommendations

- Early confirmation with an alternative assay may be obtained in all instances when an ADAMTS13 activity <20 IU/dL is not consistent with pre-test probability and clinical assessment. (2C)
- Laboratories and clinicians must have a clear understanding of the expected ranges in iTTP and cTTP at presentation of the ADAMTS13 activity assay in use. (1D)
- There must be clear clinical pathways for dealing with both normal and abnormal ADAMTS13 activity results. (1D)

TESTING FOR ANTIBODIES AGAINST ADAMTS13

When ADAMTS13 activity is <10 IU/dL at presentation, an assay for an anti-ADAMTS13 antibody is essential. If the ADAMTS13 activity is between 10 and 20 IU/dL but there is a strong clinical suspicion of TTP, an anti-ADAMTS13 antibody assay may clarify the diagnosis. The presence of anti-ADAMTS13 immunoglobulin (Ig) G antibodies supports a diagnosis of iTTP, whereas their absence may indicate a diagnosis of cTTP.

In iTTP, higher anti-ADAMTS13 antibody levels have been associated with increased mortality,[45] longer hospitalisation, an increased burden of care and more PEX sessions to achieve clinical response.[46] Identification of anti-ADAMTS13 antibodies is also useful in the management of patients and for guiding immunomodulation therapy.[1]

Patients with cTTP are treated with plasma-based therapy and, more recently, recombinant human (r)ADAMTS13. Laboratories may be required to perform these assays to identify the development of anti-ADAMTS13 or anti-rADAMTS13 antibodies.

Anti-ADAMTS13 antibody assays should be performed within 72 h of presentation when ADAMTS13 activity is <10 IU/dL.[1]

Anti-ADAMTS13 antibody assays

Autoantibodies against ADAMTS13 are neutralising IgG-antibodies or non-neutralising antibodies, which accelerate ADAMTS13 clearance from plasma. Both neutralising and non-neutralising autoantibodies can be present simultaneously in patients with iTTP.[47,48]

Apart from the analytical performance data from external quality assessment schemes for ADAMTS13 assays,[49,50] a thorough comparison between different methodologies has not been evaluated to date. There is currently no international standard or reference material for anti-ADAMTS13 antibodies: variability between assays and cut-off values to define positivity remain subjective. Refer to Table 2.

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Anti-ADAMTS13 antibody ELISA

In the United Kingdom, the majority of testing sites use the Technozym ADAMTS13 INH ELISA (Technoclone GmBH, Vienna, Austria), where results are reported in arbitrary units per mL (AU/mL). Some use an in-house ELISA laboratory-developed test (LDT). Both use full-length recombinant ADAMTS13 to capture and detect neutralising and non-neutralising anti-ADAMTS13 antibodies.[51]

Non-specific binding can occur occasionally in all ELISA methods, particularly in patients with recent or ongoing infection,[52] leading to false-positive results and reduced specificity. In the anti-ADAMTS13 antibody ELISA this is minimised by a 1/100 plasma/buffer dilution. In cases where anti-ADAMTS13 antibodies are detected but not expected, non-specific binding should be suspected: testing should be repeated at higher dilutions or using alternative buffers.

Data from the UK TTP registry[45] found that 21/273 (7.7%) of patients diagnosed with iTTP had no detectable anti-ADAMTS13 antibodies at presentation. The formation of circulating immune complexes of anti-ADAMTS13 antibodies bound to ADAMTS13 in patients with TTP during acute episodes and in remission[53] is a likely contributor to this observation. IgM or IgA anti-ADAMTS13 isotypes may also be present,[54] although the clinical significance of these remains largely unknown. Thus, clinical interpretation is crucial in interpreting anti-ADAMTS13 antibody results.

Bethesda-style inhibitor assay

A Bethesda-style inhibitor assay may be used to detect anti-ADAMTS13 antibodies, but this method lacks standardisation and only detects neutralising anti-ADAMTS13 antibodies.[55] Attempts are underway to standardise assay variables including for heat treatment, incubation times, buffers for sample dilutions and source (and potentially pH) of normal (or recombinant) ADAMTS13. Results for a Bethesda-style assay may also vary depending on the assay used to determine residual ADAMTS13 activity,[56] which may affect antibody identification at low titres.

A simple Bethesda-style inhibitor screen for anti-ADAMTS13 antibodies can be performed by mixing patient and normal plasma in a 1:1 ratio and measuring the residual ADAMTS13 activity in the mixed plasma after a short incubation.

Recommendations

- Anti-ADAMTS13 antibody assays for clinical use should only be performed in laboratories of regional TTP centres. (1C)
- Anti-ADAMTS13 antibody assays must be performed when ADAMTS13 activity <10 IU/dL, within 72 h of presentation. (1B)
- Anti-ADAMTS13 antibody assays may be performed when there is strong clinical suspicion of TTP but ADAMTS13 activity is between 10 and 20 IU/dL. (2B)
- Anti-ADAMTS13 antibody assays may be repeated when ADAMTS13 activity levels remain low despite treatment for acute iTTP to guide clinical decision making for immunosuppression. (2C)
- Anti-ADAMTS13 antibody assays may be used to exclude acquired inhibitors for those receiving recombinant ADAMTS13. (2B)
- Anti-ADAMTS13 antibody assays may be performed in patients with suspected relapse of iTTP. (2B)
- An ELISA is recommended over non-standardised Bethesda-style inhibitor assay or mixing studies for the detection of anti-ADAMTS13 antibodies. (2D)
- Genomic analysis of the *ADAMTS13* gene is required to confirm *c*TTP. (1C)

TESTING FOR ADAMTS13 ANTIGEN

The clinical utility of ADAMTS13 antigen levels appears to be limited in routine clinical practice, and there is insufficient evidence to recommend them outside of research settings.

More recently it has been shown that caplacizumab therapy might be safely discontinued when ADAMTS13 antigen is >30 IU/dL in patients who have a persistently low ADAMTS13 activity (<10 IU/dL) during prolonged treatment.[57]

Therefore, although ADAMTS13 antigen assays are not recommended for routine clinical use, they may be considered in individual patients for predicting exacerbation or relapse off-treatment when ADAMTS13 activity remains low.

ADAMTS13 antigen is measured by ELISA, using either a commercial kit or a LDT; results are not comparable between assays. Refer to Table 1c.

Recommendations

- ADAMTS13 antigen assays are not recommended for routine clinical use. (2C)
- ADAMTS13 antigen assays may be considered in individual patients on a case-by-case basis. (2C)
- ADAMTS13 antigen assays for clinical use should only be performed in laboratories of regional TTP centres. (1C)

ADAMTS13 CONFORMATION (OPEN/CLOSED INDEX) ASSAY

An area of recent interest is the ADAMTS13 Conformation ELISA. ADAMTS13 conformational changes, based on the presence of ADAMTS13 IgG antibody, may provide additional information for patients in the remission phase or

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in relapse. The open ADAMTS13 conformation appears to occur before a reduction in ADAMTS13 activity, rendering patients at risk of relapsing and requiring closer monitoring.[58] Quantitation of ADAMTS13 antigen is required to determine the significance of any conformational change.

These assays may be helpful where activity is borderline (10–20 IU/dL).[59] In remission, an open conformation index predicted increased relapse risk even when activity was normalised.[60] When assessed at peak ADAMTS13 activity (>60%), patients with an open conformation had markedly higher relapse rates than those with a closed conformation (1-year 46% vs. 9%, 2-year 62% vs. 23%).[58] Open conformation could precede relapse and persist despite falling anti-ADAMTS13 antibody levels, indicating potential value of these assays. Assay availability is likely to remain confined to regional TTP centres; conformation testing should be viewed as a complement to routine monitoring rather than a first-line tool.[58–60]

Recommendations

- ADAMTS13 conformation assays are not currently recommended for routine clinical use. (2C)
- ADAMTS13 conformation assays for clinical use should only be performed in laboratories of regional TTP centres. (1C)

LABORATORY MONITORING IN RECOVERY AND RELAPSE

Longitudinal ADAMTS13 activity measurement is the cornerstone of post-diagnosis surveillance in both iTTP and cTTP. However, optimal testing frequency and thresholds for intervention remain incompletely defined. Updated outcome definitions from the International Working Group distinguish clinical remission/relapse from ADAMTS13 activity remission/relapse, embedding laboratory results within disease classification and recognising the impact of therapies such as caplacizumab.[61]

Monitoring on a single platform permits sensitive detection of decreasing ADAMTS13 activity in disease relapse whereas using different platforms may introduce variability not correlated to the clinical status of the patient. Laboratories should consider confirmation on a second platform when an assay is unexpectedly low or results appear discordant with the clinical picture.[33,34]

VWF activity is expected to reduce following treatment with caplacizumab in iTTP, and measurement of VWF activity may be considered as a biomarker for drug monitoring.[62] However, the effect of caplacizumab on VWF activity is assay dependent: VWF:GPIbM,[26,62] VWF:RCo and VWF:GPIbR[26,63] assays may be expected to be very low post-treatment, but VWF:CB, VWF:Ab and VWF:Ag assays may be much less affected.[26,63]

Recommendations

- ADAMTS13 activity assay differences indicate that for monitoring, and identification of remission and relapse, the same analytical method should be used in any single patient to retain sensitivity for clinically significant changes. (1C)
- Laboratories may consider confirmation on a second platform when ADAMTS13 activity is unexpectedly low, or results appear discordant with the clinical picture. (2B)

EXTERNAL QUALITY ASSURANCE

All laboratories that perform assays for ADAMTS13 (activity, anti-ADAMTS13 antibody or antigen) should participate in external quality assurance/proficiency testing. In the United Kingdom, laboratories can subscribe through UK National External Quality Assurance Scheme (NEQAS) for Blood Coagulation (www.ukneqasbc.org) or through External quality Control for Assays and Tests (ECAT) with a focus on Thrombosis and Haemostasis (www.ecat.nl).

CLINICAL SUPPORT FOR LABORATORY PRACTICE

Given the complexities of ADAMTS13 testing, alongside the clinical risks associated with delays in any aspect of the management pathway, support of the laboratory from clinical haematologists is essential. Decisions regarding treatment for patients with strong clinical suspicion of TTP must be made by critical care and haematology teams without waiting for laboratory results. Teams involved in the care of such patients should be aware of the agreed turnaround times for ADAMTS13 testing so that laboratory scientists are protected from clinical time pressures to allow safe processing of samples and reporting of results.

Benefits of job-planned time for a haematology consultant to offer laboratory support may include:

- Involvement of laboratory scientists in multidisciplinary team meetings for TMA (including TTP);
- A gatekeeping role, ensuring the appropriate pre-test risk assessment before sending and/or testing a sample;
- Discussing and reporting results with the laboratory team;
- Developing and updating local policies and procedures for: blood film review and escalation to a TTP/TMA pathway; reporting of haematology and biochemistry results in the presence of in vivo haemolysis; TTP tests and testing, including new developments in the TMA/TTP fields;
- Providing clinical advice regarding pre-test probability scores alongside laboratory results.

Without appropriate support from clinicians with interest and experience in TTP, ADAMTS13 testing carries

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significant risks of both over and late diagnosis, with consequent resource and safety ramifications.

AUTHOR CONTRIBUTIONS

Thomas Pitchford: Investigation; methodology; validation; writing – review and editing. **Richard Gooding:** Methodology; validation; writing – review and editing. **Sean Platton:** Writing – original draft; methodology; validation; writing – review and editing; supervision; project administration. **Rebekah Fretwell:** Methodology; writing – review and editing; validation. **Caroline Lawrence:** Methodology; validation; writing – review and editing; investigation. **P. Vince Jenkins:** Methodology; validation; writing – review and editing; investigation. **Stephen MacDonald:** Supervision; writing – review and editing; validation; methodology; investigation. **Deepak Singh:** Investigation; methodology; validation; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

The BSH paid the expenses incurred during the writing of this guidance. All authors have made a declaration of interests to the BSH, which may be viewed on request. The following authors have declared reimbursement for travel, speaking or consultancy fees: SM—Werfen and Randox; TP—Sysmex; RG—Roche and Sobi; SP—Sysmex, Pfizer and Roche-Chugai. The following members of the writing group list initials have no conflicts of interest to declare: DS; RF; PVJ; CL.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

REVIEW PROCESS

Members of the writing group will inform the writing group chair if any new evidence becomes available that would alter the strength of the recommendations made in this document or render it obsolete. The document will be reviewed regularly by the relevant task force to search for any new evidence. The document will be archived and removed from the BSH current guidelines website if it becomes obsolete. Please check the BSH guidelines website (www.b-s-h.org.uk/guidelines) for any addenda that may be produced after the initial publication.

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